Phase 1: Modelling, Equilibria and Linearization of the Acrobot

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Abstract. We verify the given Acrobot model, find its zero-torque equilibria, linearize at each, and derive the transfer functions from the elbow torque τ to both joint angles.

I. Nonlinear Model

A. System and Parameters. Two-link pendulum (Fig. 1) with a passive shoulder and a motor at the elbow [1], [2].

- Angles: q_1 = link 1 from the downward vertical; q_2 = link 2 relative to link 1; $\varphi = q_1 + q_2$.
- State: $x = [q_1,\ q_2,\ \dot{q}_1,\ \dot{q}_2]^T$.
- Input: elbow motor torque τ, $|\tau| \le 20\,\mathrm{N\,m}$, the only quantity we command ($\tau = 0$ in this phase; set by a controller in later phases).
- $m_1 = m_2 = 1$ kg, $l_1 = 1$ m, $l_2 = 2$ m, $l_{c1} = 0.5$ m, $l_{c2} = 1$ m, $I_1 = 0.083$, $I_2 = 0.33\ \mathrm{kg\,m^2}$, $g = 9.81\ \mathrm{m/s^2}$.

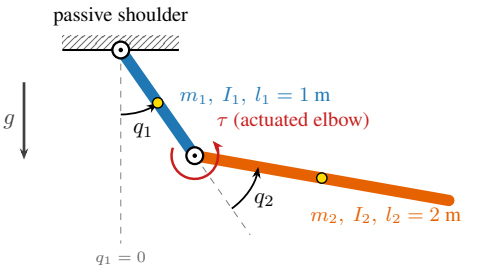


Fig. 1. Acrobot schematic (to scale); dots mark the centres of mass.

B. Given Equations of Motion.

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} = \tau_g(q) + B\tau, \qquad B = [0,\ 1]^T \tag{1}$$

(B applies τ to the elbow equation only; $c_2 = \cos q_2$, $s_2 = \sin q_2$.)

C. Derivation. *1) Potential energy and gravity torque:*

$$U = -(m_1 l_{c1} + m_2 l_1) g \cos q_1 - m_2 l_{c2} g \cos\varphi = -14.715\cos q_1 - 9.81\cos\varphi, \qquad \tau_g = -\frac{\partial U}{\partial q} = -9.81\begin{bmatrix} 1.5\sin q_1 + \sin\varphi \\ \sin\varphi \end{bmatrix}. \tag{2}$$

2) Kinetic energy and inertia matrix: the velocity of centre 2 is the elbow velocity ($l_1\dot{q}_1$) plus the rotation about the elbow ($l_{c2}\dot{\varphi}$), at angle q_2 apart (law of cosines):

$$T = \tfrac{1}{2}(m_1 l_{c1}^2 + I_1)\dot{q}_1^2 + \tfrac{1}{2} I_2 \dot{\varphi}^2 + \tfrac{1}{2} m_2 \left(l_1^2\dot{q}_1^2 + l_{c2}^2\dot{\varphi}^2 + 2 l_1 l_{c2} c_2 \dot{q}_1\dot{\varphi}\right) = \tfrac{1}{2}\dot{q}^T M \dot{q}, \quad M(q) = \begin{bmatrix} 2.663 + 2c_2 & 1.33 + c_2 \\ 1.33 + c_2 & 1.33 \end{bmatrix}. \tag{3}$$

3) Coriolis term (Euler–Lagrange): with $\partial T/\partial \dot{q} = M\dot{q}$ and $\dot{M} = \frac{\partial M}{\partial q_2}\dot{q}_2 = -s_2\dot{q}_2 \left[\begin{smallmatrix} 2 & 1 \\ 1 & 0 \end{smallmatrix}\right]$,

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}} - \frac{\partial T}{\partial q} = \tau_g + B\tau \ \Rightarrow\ M\ddot{q} + \underbrace{\left(\dot{M}\dot{q} - \frac{\partial T}{\partial q}\right)}_{=C\dot{q}} = \tau_g + B\tau, \qquad C(q,\dot{q}) = s_2\begin{bmatrix} -2\dot{q}_2 & -\dot{q}_2 \\ \dot{q}_1 & 0 \end{bmatrix}. \tag{4}$$

These are the same τ_g, M and C as in the given model.

D. State-Space Form. M is always invertible, so solving (1) for $\ddot{q}$ gives the state equation:

$$\det M = (2.663 + 2c_2)(1.33) - (1.33 + c_2)^2 = 1.7729 - c_2^2 \ge 0.7729 > 0 \ \ (c_2^2 \le 1), \qquad \dot{x} = f(x,\tau) = \begin{bmatrix} \dot{q} \\ M^{-1}\left(\tau_g + B\tau - C\dot{q}\right) \end{bmatrix}. \tag{5}$$

II. Equilibria

A. Conditions. At rest $\dot{q} = \ddot{q} = 0$, and the input torque is given as $\tau^* = 0$. Then (1) gives $\tau_g(q^*) = 0$:

$$\text{row 2: } \sin(q_1^* + q_2^*) = 0, \qquad \text{row 1: } 1.5\sin q_1^* + \underbrace{\sin(q_1^* + q_2^*)}_{=0} = 0 \ \Rightarrow\ \sin q_1^* = 0.$$

B. Solutions. $q_1^* \in \{0,\pi\}$, $q_1^* + q_2^* \in \{0,\pi\} \Rightarrow q_2^* \in \{0,\pi\}$ (mod 2π): **four equilibria**, all with both links vertical (Table I).

Table I. Equilibria for $\tau^* = 0$

	Posture	(q_1^*, q_2^*)	U [J]	Description
E_1	down–down	$(0,0)$	-24.525	both links hanging
E_2	down–up	$(0,\pi)$	-4.905	link 1 down, link 2 up
E_3	up–up	$(\pi,0)$	$+24.525$	fully upright (target)
E_4	up–down	(π,π)	$+4.905$	link 1 up, link 2 down

III. Linearization

A. Deviation Variables. $\tilde{q} = [q_1 - q_1^*,\ q_2 - q_2^*]^T$, $\tilde{x} = [\tilde{q}; \dot{\tilde{q}}] = x - x^*$, $\tilde{\tau} = \tau - \tau^* = \tau$. First-order Taylor expansion of (5) about $(x^*, 0)$, using $f(x^*, 0) = 0$: $\dot{\tilde{x}} \approx \frac{\partial f}{\partial x}\big|_* \tilde{x} + \frac{\partial f}{\partial \tau}\big|_* \tilde{\tau}$.

B. Jacobians. *Top half:* $\frac{d}{dt}\tilde{q} = \dot{\tilde{q}} \Rightarrow$ rows $[0\ \ I_2]$. *Bottom half:* $h = M^{-1}(\tau_g + B\tau - C\dot{q})$; at $(x^*, 0)$, $\partial(C\dot{q})/\partial\dot{q} = 0$ ($C\dot{q}$ is quadratic in $\dot{q}$, $\dot{q}^* = 0$) and $(\partial M^{-1}/\partial q)(\tau_g + B\tau) = 0$ ($\tau_g(q^*) = 0$, $\tau^* = 0$). Hence

$$\ddot{\tilde{q}} = P\tilde{q} + r\tilde{\tau}, \quad P = M(q^*)^{-1}K(q^*), \quad r = M(q^*)^{-1}B, \qquad K(q^*) = \frac{\partial \tau_g}{\partial q}\Big|_{q^*} = -9.81 \begin{bmatrix} 1.5\cos q_1^* + \cos\varphi^* & \cos\varphi^* \\ \cos\varphi^* & \cos\varphi^* \end{bmatrix}. \tag{6}$$

C. Linear State-Space Model. Stacking both halves, with the joint angles as outputs, $y = \tilde{q} = [q_1 - q_1^*,\ q_2 - q_2^*]^T$:

$$\boxed{\dot{\tilde{x}} = A\tilde{x} + B_\ell \tilde{\tau}, \quad y = C_y\tilde{x} + D\tilde{\tau}, \qquad A = \begin{bmatrix} 0 & I_2 \\ P & 0 \end{bmatrix},\ B_\ell = \begin{bmatrix} 0 \\ r \end{bmatrix},\ C_y = \begin{bmatrix} I_2 & 0 \end{bmatrix},\ D = 0} \tag{7}$$

D. Example: $E_1 = (0,0)$. $\cos q_1^* = \cos\varphi^* = c_2 = 1$:

$$K = \begin{bmatrix} -24.525 & -9.81 \\ -9.81 & -9.81 \end{bmatrix}, \quad M = \begin{bmatrix} 4.663 & 2.33 \\ 2.33 & 1.33 \end{bmatrix}, \quad \det M = 0.7729, \quad M^{-1} = \frac{1}{0.7729}\begin{bmatrix} 1.33 & -2.33 \\ -2.33 & 4.663 \end{bmatrix},$$

$$P_{E1} = M^{-1}K = \begin{bmatrix} -12.6292 & 12.6926 \\ 14.7488 & -29.6119 \end{bmatrix}, \qquad r_{E1} = \begin{bmatrix} -3.0147 \\ 6.0332 \end{bmatrix}.$$

E. All Equilibria. M depends only on q_2^*; K only on the signs of $\cos q_1^*$, $\cos\varphi^*$. Upside down, gravity flips: $K_{E3} = -K_{E1}$ with $M_{E3} = M_{E1}$, so $P_{E3} = -P_{E1}$; likewise $P_{E4} = -P_{E2}$. Results in Table II; with (7) they give (A, B_ℓ, C_y, D) at each equilibrium.

Table II. Linearization results ($\det M = 0.7729$ at all four)

	$(\cos q_1^*, \cos\varphi^*)$	$M(q^*)$	$P = M^{-1}K$	$r = M^{-1}B$
E_1	$(1,1)$	$\left[\begin{smallmatrix} 4.663 & 2.33 \\ 2.33 & 1.33 \end{smallmatrix}\right]$	$\left[\begin{smallmatrix} -12.6292 & 12.6926 \\ 14.7488 & -29.6119 \end{smallmatrix}\right]$	$\left[\begin{smallmatrix} -3.0147 \\ 6.0332 \end{smallmatrix}\right]$
E_2	$(1,-1)$	$\left[\begin{smallmatrix} 0.663 & 0.33 \\ 0.33 & 1.33 \end{smallmatrix}\right]$	$\left[\begin{smallmatrix} -12.6292 & 12.6926 \\ 10.5095 & 4.2266 \end{smallmatrix}\right]$	$\left[\begin{smallmatrix} -0.4270 \\ 0.8578 \end{smallmatrix}\right]$
E_3	$(-1,-1)$	as E_1	$-P_{E1}$	r_{E1}
E_4	$(-1,1)$	as E_2	$-P_{E2}$	r_{E2}

IV. Transfer Functions

A. Laplace Transform. With $\tilde{Q}_i(s) = \mathscr{L}\{\tilde{q}_i(t)\}$, $\tilde{\mathscr{T}}(s) = \mathscr{L}\{\tilde{\tau}(t)\}$, $\tilde{Q} = [\tilde{Q}_1, \tilde{Q}_2]^T$ and zero initial conditions ($\ddot{\tilde{q}} \to s^2\tilde{Q}$):

$$s^2\tilde{Q} = P\tilde{Q} + r\tilde{\mathscr{T}} \ \Rightarrow\ (s^2 I - P)\,\tilde{Q}(s) = r\,\tilde{\mathscr{T}}(s), \qquad (s^2 I - P)^{-1} = \frac{1}{\Delta(s)}\begin{bmatrix} s^2 - P_{22} & P_{12} \\ P_{21} & s^2 - P_{11} \end{bmatrix}, \quad \Delta(s) = s^4 - \mathrm{tr}(P)\,s^2 + \det P.$$

B. General Formula. Multiplying by r and dividing each row by $\tilde{\mathscr{T}}$ ($G_i = \tilde{Q}_i/\tilde{\mathscr{T}}$):

$$\boxed{G_1(s) = \frac{r_1 s^2 + (P_{12}r_2 - P_{22}r_1)}{\Delta(s)}, \qquad G_2(s) = \frac{r_2 s^2 + (P_{21}r_1 - P_{11}r_2)}{\Delta(s)}} \tag{8}$$

Poles: roots of Δ (= eigenvalues of A); with $\lambda = s^2$, $\lambda^2 - \mathrm{tr}(P)\lambda + \det P = 0$, then $s = \pm\sqrt{\lambda}$. *Zeros:* $s^2 = -(P_{12}r_2 - P_{22}r_1)/r_1$ for G_1 and $s^2 = -(P_{21}r_1 - P_{11}r_2)/r_2$ for G_2.

C. Example: E_1. $\operatorname{tr} P = -42.2410$, $\det P = 186.7719 \Rightarrow \lambda = s^2 = -5.018, -37.224 \Rightarrow$ poles $\pm j2.2400$, $\pm j6.1011$; numerator constants $P_{12}r_2 - P_{22}r_1 = -12.6926$, $P_{21}r_1 - P_{11}r_2 = 31.7316 \Rightarrow G_1 = -3.0147\,(s^2+4.2103)/\Delta$, $G_2 = 6.0332\,(s^2+5.2595)/\Delta$ (row E_1 of Table III).

Table III. Transfer functions $G_i(s) = N_i(s)/\Delta(s)$ at the four equilibria

	Denominator $\Delta(s)$ (monic)	$N_1(s)$ $(\tau \to q_1)$	$N_2(s)$ $(\tau \to q_2)$	Poles	Zeros of G_1	Zeros of G_2
E_1	$s^4 + 42.2410\,s^2 + 186.7719$	$-3.0147\,(s^2+4.2103)$	$6.0332\,(s^2+5.2595)$	$\pm j2.2400,\ \pm j6.1011$	$\pm j2.0519$	$\pm j2.2934$
E_2	$s^4 + 8.4025\,s^2 - 186.7719$	$-0.4270\,(s^2-29.7273)$	$0.8578\,(s^2+7.3982)$	$\pm 3.1775,\ \pm j4.3010$	± 5.4523	$\pm j2.7200$
E_3	$s^4 - 42.2410\,s^2 + 186.7719$	$-3.0147\,(s^2-4.2103)$	$6.0332\,(s^2-5.2595)$	$\pm 2.2400,\ \pm 6.1011$	± 2.0519	± 2.2934
E_4	$s^4 - 8.4025\,s^2 - 186.7719$	$-0.4270\,(s^2+29.7273)$	$0.8578\,(s^2-7.3982)$	$\pm 4.3010,\ \pm j3.1775$	$\pm j5.4523$	± 7.2700

D. All Equilibria. Table III. With no friction only even powers of s occur, so poles and zeros come in pairs $\pm p$; since $P_{E3} = -P_{E1}$, each $\pm j\omega$ at E_1 becomes $\pm\omega$ at E_3 (likewise E_2/E_4).

V. Verification

- The model derivation and all values in Tables II and III were verified in MATLAB (agreement to four decimals).
- Nonlinear (5) vs. linear (7), simulated in Python from the same small deviation, $\tau = 0$ (Fig. 2): near E_1 from $\tilde{q}(0) = (0.1, -0.1)$ rad they agree within 1.0° (q_1) and 2.0° (q_2) over 8 s; near E_3 from $\tilde{q}_1(0) = 10^{-3}$ rad the error is 0.02% at $|\tilde{q}_1| = 1^\circ$.
- With $\tau = 0$, $T + U$ stays constant to 10^{-8} (no friction).
- **Animation** (Manim) showing the model, the four equilibria and their responses: **watch the video**.

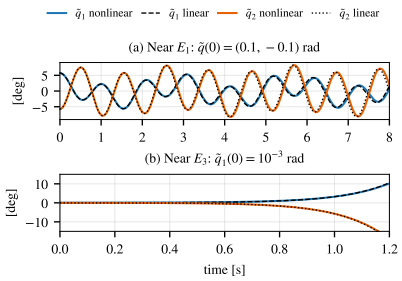


Fig. 2. Nonlinear (colour) vs. linear (black) responses near (a) E_1, (b) E_3 (Python).

References. [1] M. W. Spong, "The swing up control problem for the Acrobot," *IEEE Control Syst. Mag.*, vol. 15, no. 1, pp. 49–55, 1995. [2] R. Tedrake, *Underactuated Robotics*, MIT, 2024. [Online]. Available: https://underactuated.csail.mit.edu